DATA SCIENCE

DATA VISUALIZATION

INTRODUCTION

Data Science is the process that *transforms enormous data* to *certain knowledge presentation*, in which a human being finds a meaning.

- The *massive complex data* needs to be *transformed* into the *expression* that a human can understand.

One of the important roles of data science is to develop such knowledge presentation according to a purpose.

Knowledge presentation is the *framework that converts a large amount of complex data into a particular data or procedure that human being can figure out* based on an purpose.

VISUALIZATION

Visualization **is a method of computing, it** ***transforms the symbolic into the geometric*****, enabling the data scientists to observe the simulations and computations.**

- i.e. visualization is the conversation of *data* into a *visual format* so the characteristics of the data and the relationships among data items or attributes can be *analyzed or reported*.

- Data visualization is one of the most powerful and appealing techniques for data exploration
  - Human have ability to analyses large amounts of information that presented visually
  - Human can detect general pattern and trends
  - Human can detect outliers and unusual patterns

IMPORTANT OF DATA VISUALIZATION

Effective data visualization is an important aspect of data science, for at least three distinct reasons:

1) Exploratory data analysis:

- What does the data really look like? Getting a handle on what we are dealing with is the first step of any serious analysis.

- Plots and visualizations are the best way to do this.

IMPORTANT OF DATA VISUALIZATION

Effective data visualization is an important aspect of data science, for at least three distinct reasons:

2) Error detection:

- Did we do something stupid in our analysis? Feeding unvisualized data to any machine learning algorithm is asking for trouble.
  - Problems with outlier points, insufficient cleaning, and erroneous assumptions reveal themselves immediately when properly visualizing your data.

- The summary statistic hides what the model is really doing, 77.8% accurate. Taking a good hard look what we are getting right vs. wrong is the first step to performing better.

IMPORTANT OF DATA VISUALIZATION

Effective data visualization is an important aspect of data science, for at least three distinct reasons:

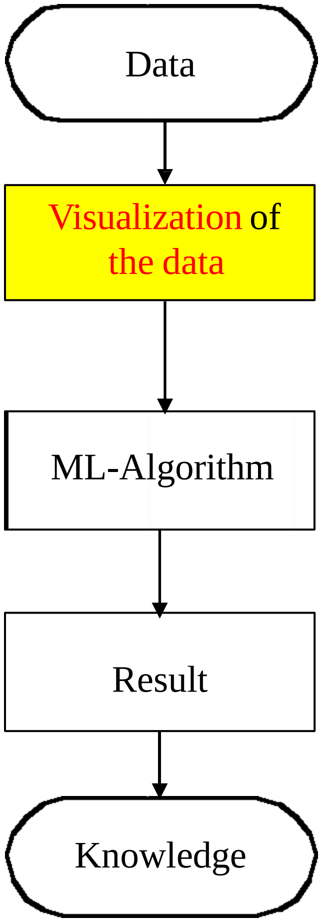
3) Communication:

- Can we present what we have learned effectively to others? Meaningful results become actionable only after they are shared. Our success as a data scientists rests on convincing other people that we know what we are talking about.

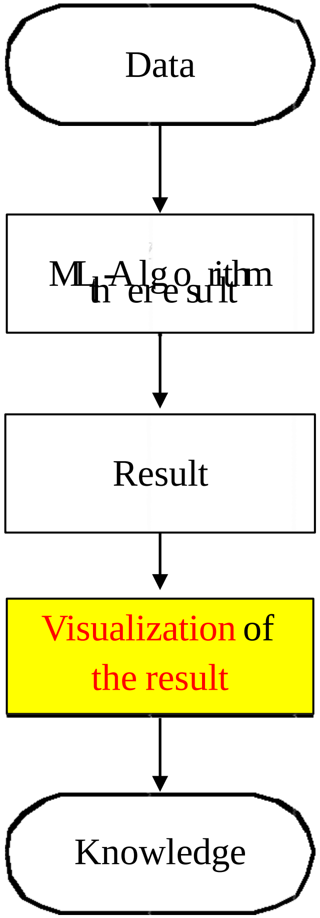
- A picture is worth 1,000 words, especially when we are giving a presentation to a skeptical audience.

VISUALIZATION TYPES

- Visualization of Data
  (preceding visualization)



- Visualization of results
  (subsequent visualization)



DATA VISUALIZATION

Data can be visualized in 1D, 2D or 3D, well-known visualization methods

One-dimension **visualizing methods:**

- Histogram
- Pie Chart

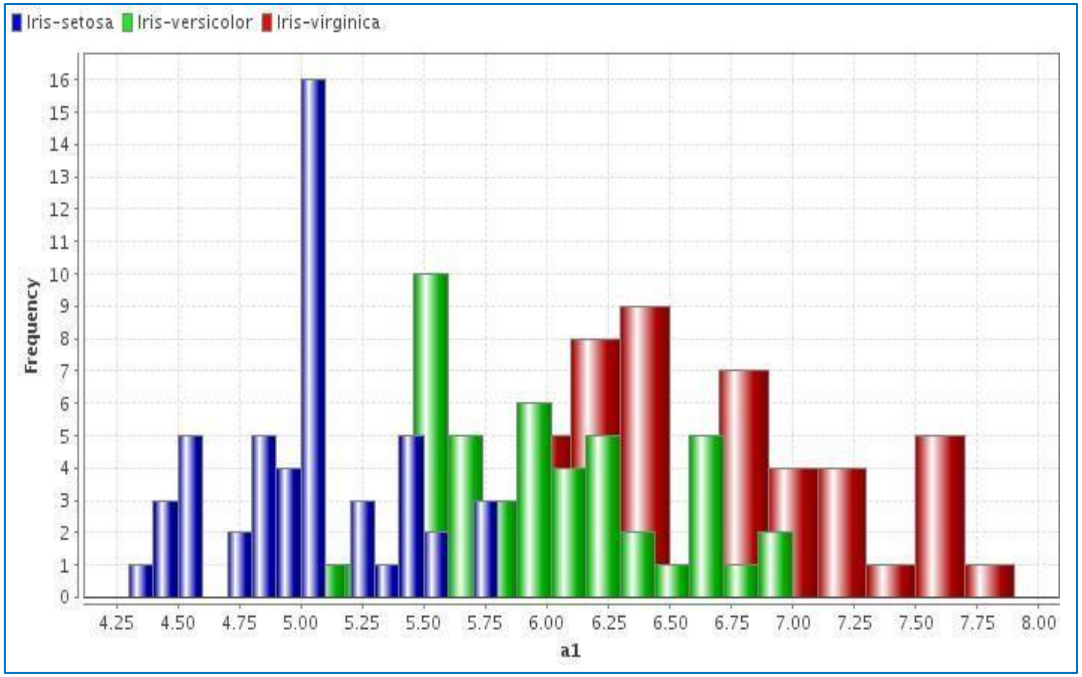
Multi-dimensions **visualizing methods:**

- Scatter plots
- Parallel coordinates

DATA VISUALIZATION

Histogram: **A histogram is the graphical representation generally employed for a continuous distribution.**

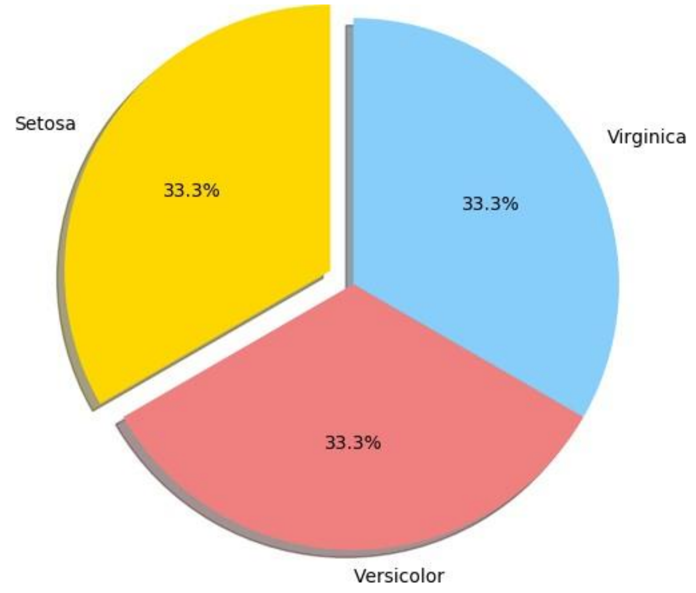
- Histogram shows the frequency of the distinct values of a ceratin column/attribute.
- A simple histogram can be a great first step in understanding a dataset.



DATA VISUALIZATION

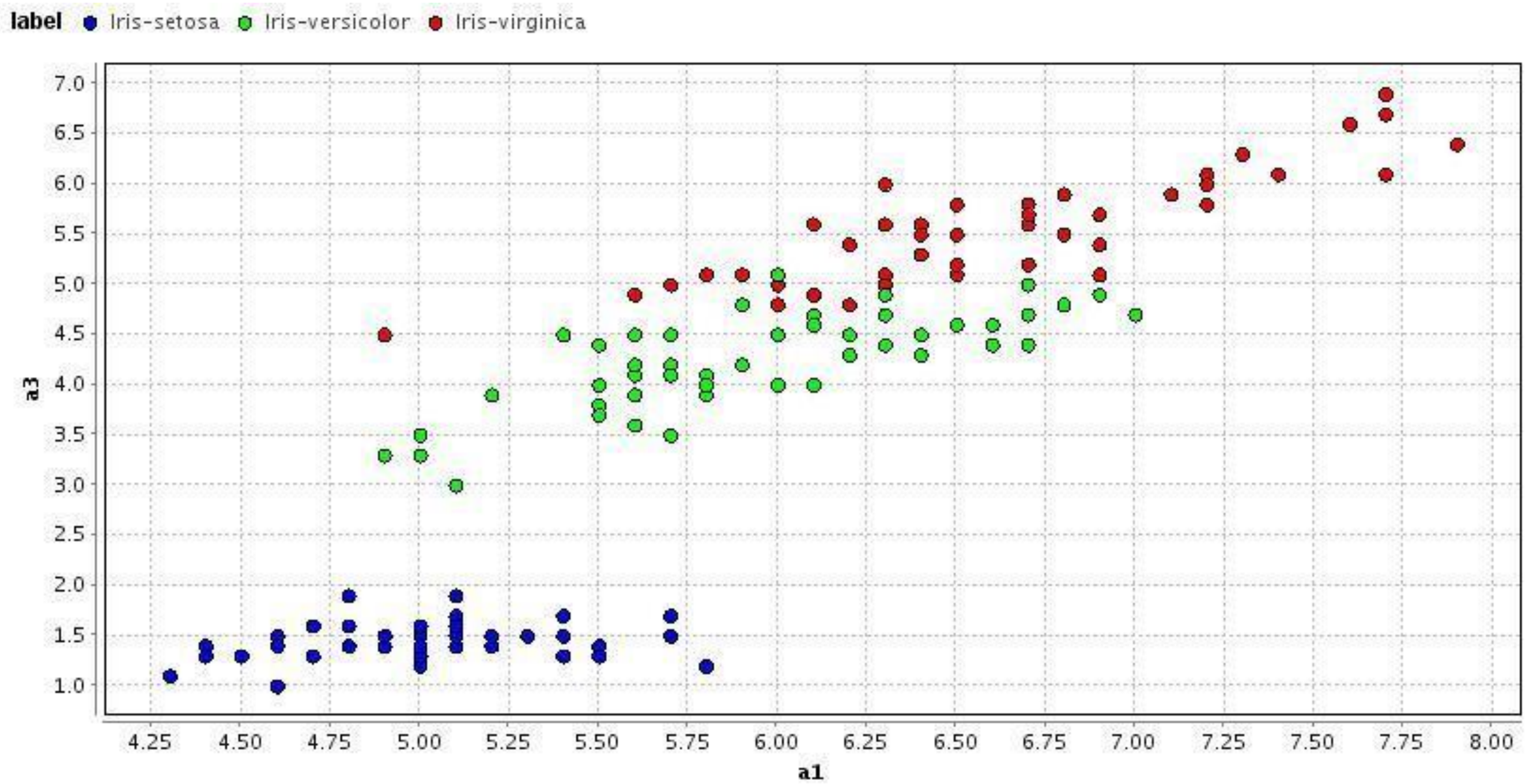
Pie chart: **A pie chart is also a pictorial representation of the uni-variate data.**

- In a pie chart or diagram a circle is drawn and it is divided into sectors on such a way that the area of the sectors are proportional to the magnitudes given.



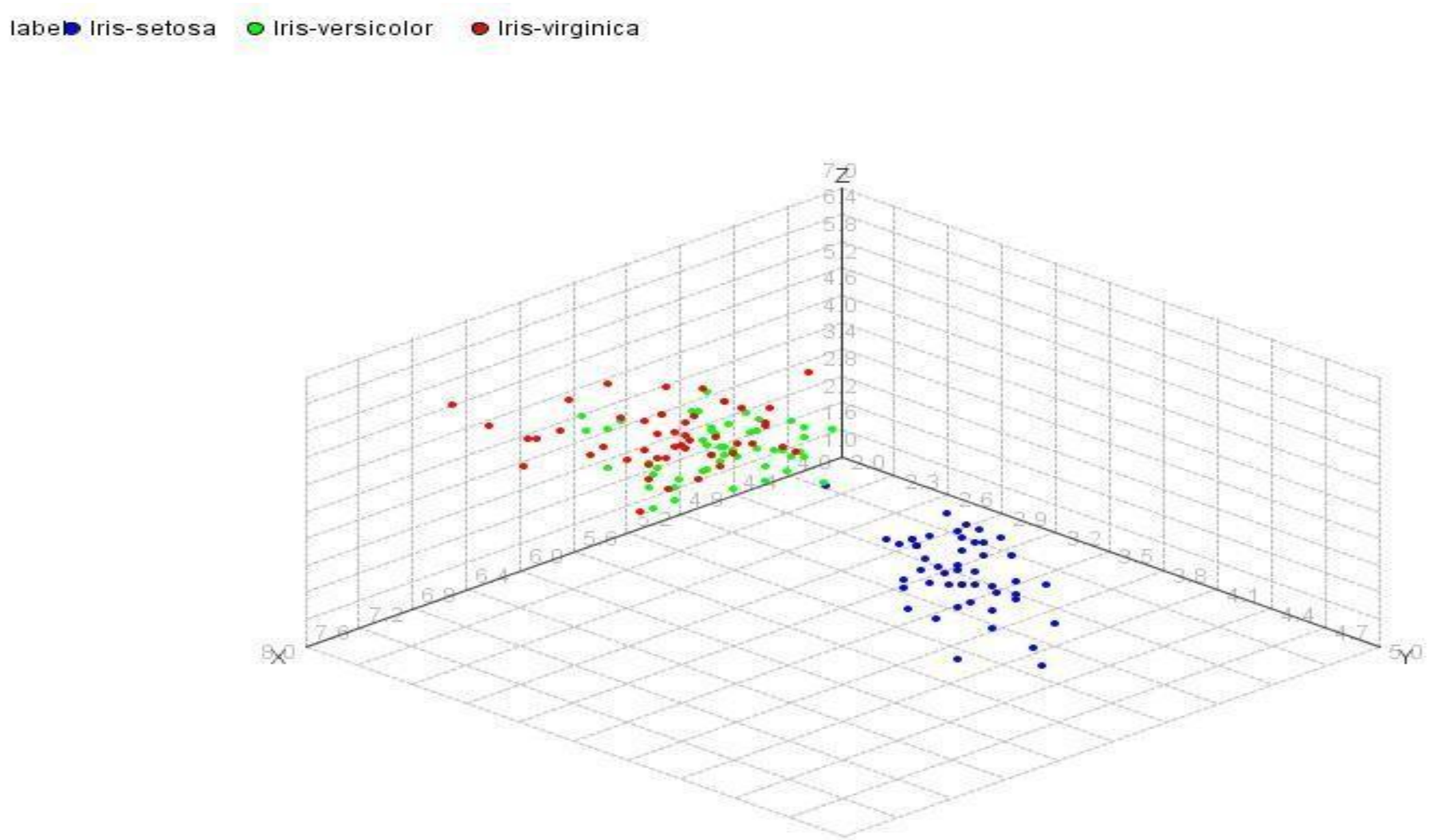
DATA VISUALIZATION

Scatter plot (2D): A number of variables are graphed to two axis.



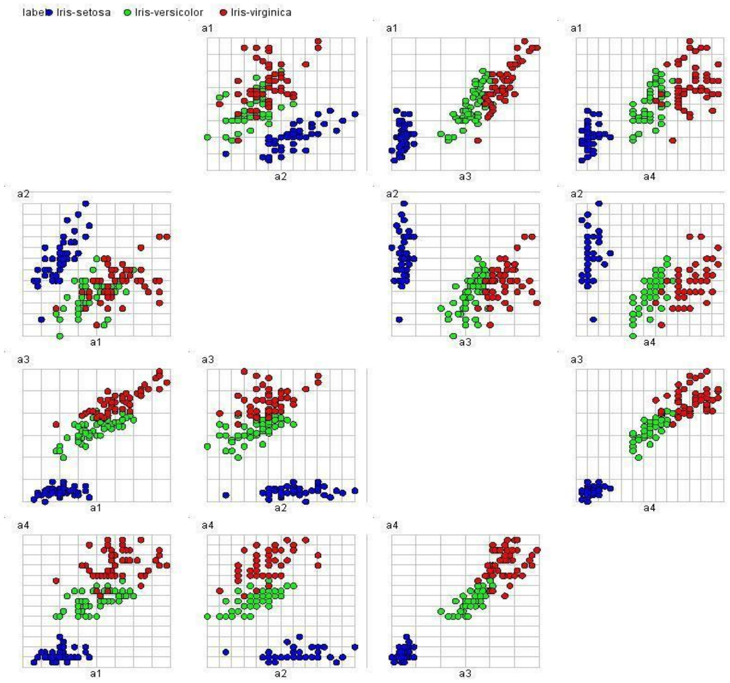
DATA VISUALIZATION

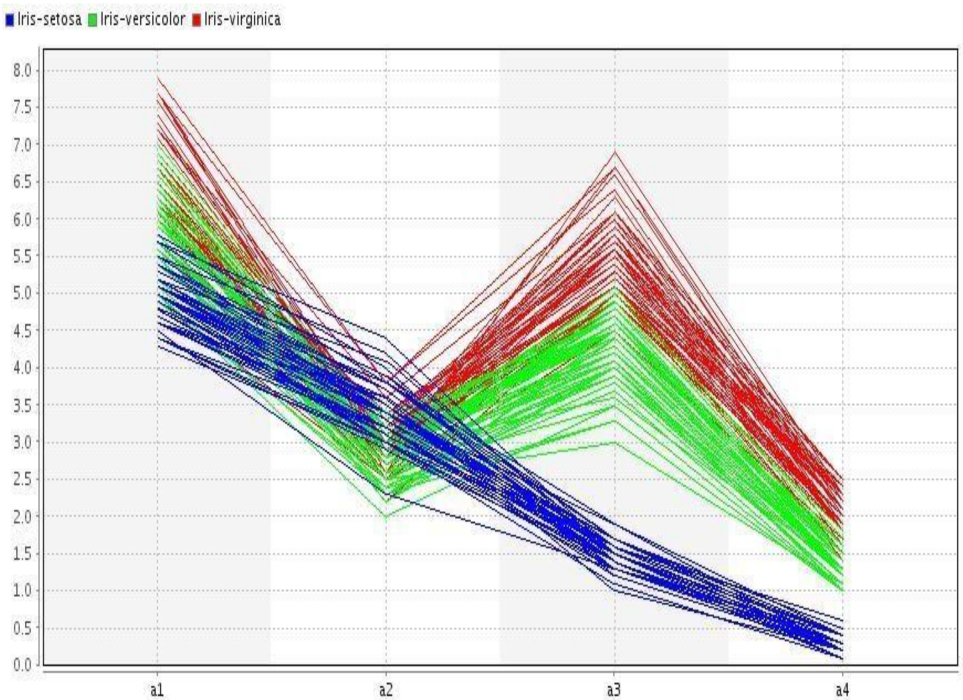
Scatter plot (3D): A number of variables are graphed to 3 axis.



DATA VISUALIZATION

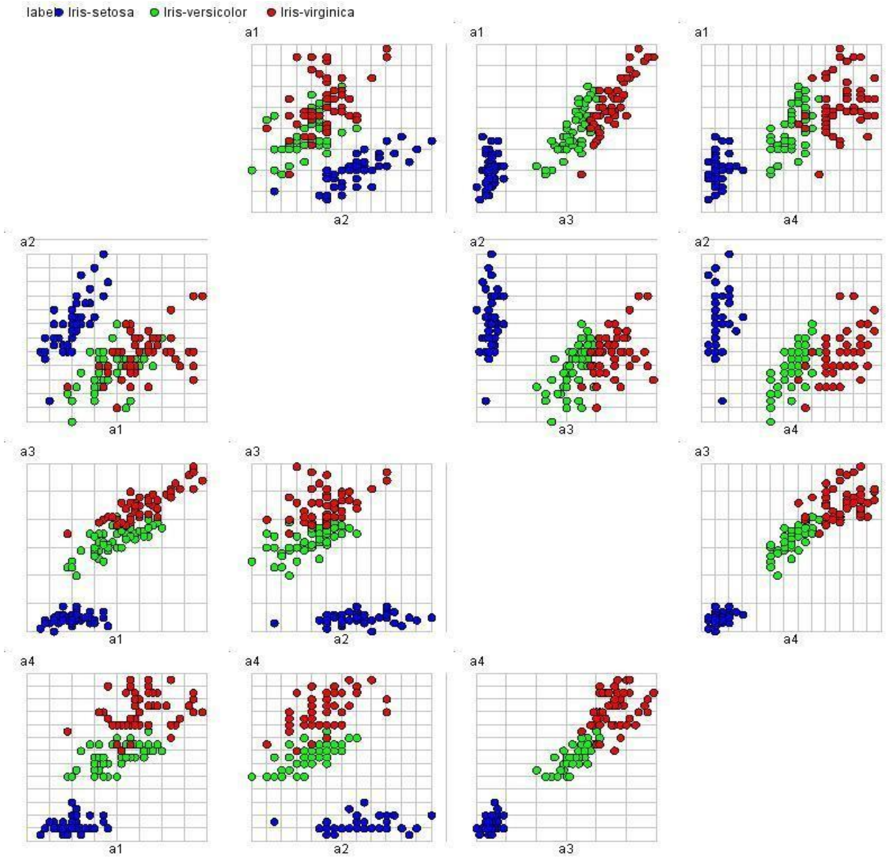
For four or more dimension visualization the Scatterplots or Parallel Coordinates can be used.





DATA VISUALIZATION

Represent each possible pair of variables in their own 2D scatterplot matrix.



DATA VISUALIZATION

Parallel Coordinates:

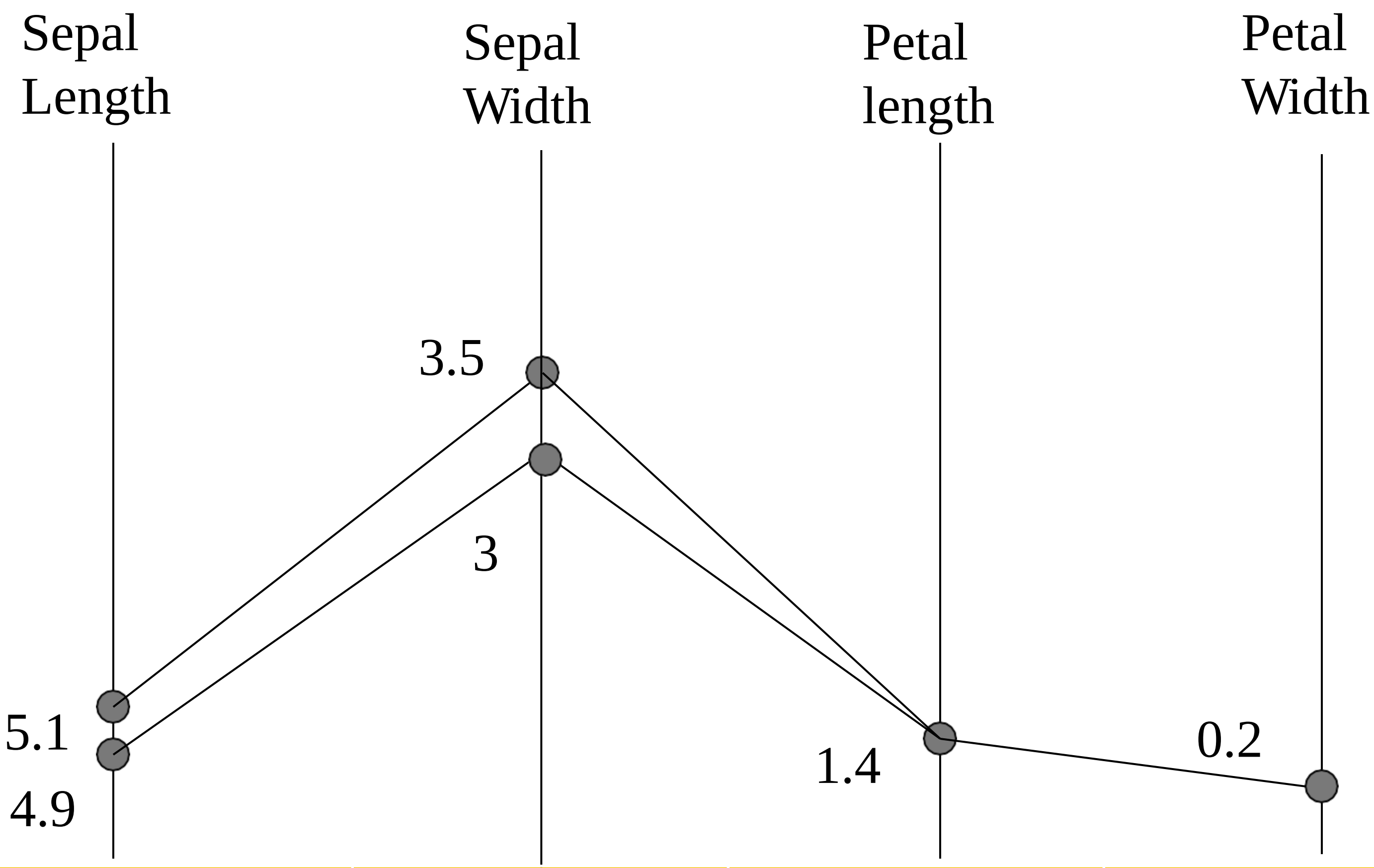
Iris setosa

sepal length	sepal width	petal length	petal width
5.1	3.5	1.4	0.2
4.9	3	1.4	0.2
...	...	...	...
5.9	3	5.1	1.8

Iris versicolor

Iris virginica

DATA VISUALIZATION



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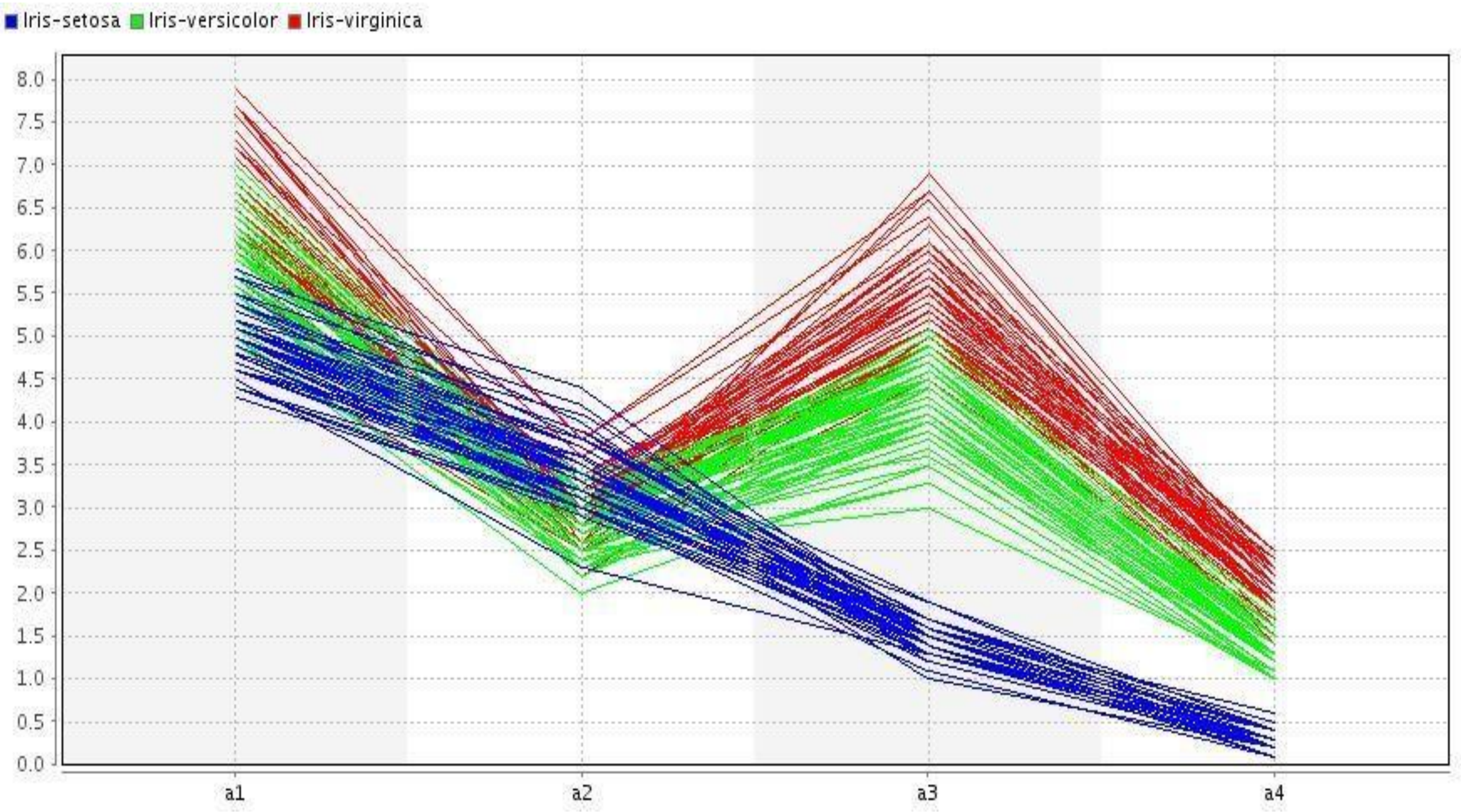
DATA VISUALIZATION



DATA VISUALIZATION

Parallel Coordinates summary:

- Each data point is a line
- Similar points correspond to similar lines
- Lines crossing over correspond to negatively correlated attributes
- ***Problems***: order of axes, limited dimensions

VISUALIZATION WEAKNESSES

Extremely difficult to find many types of high-dimensional patterns

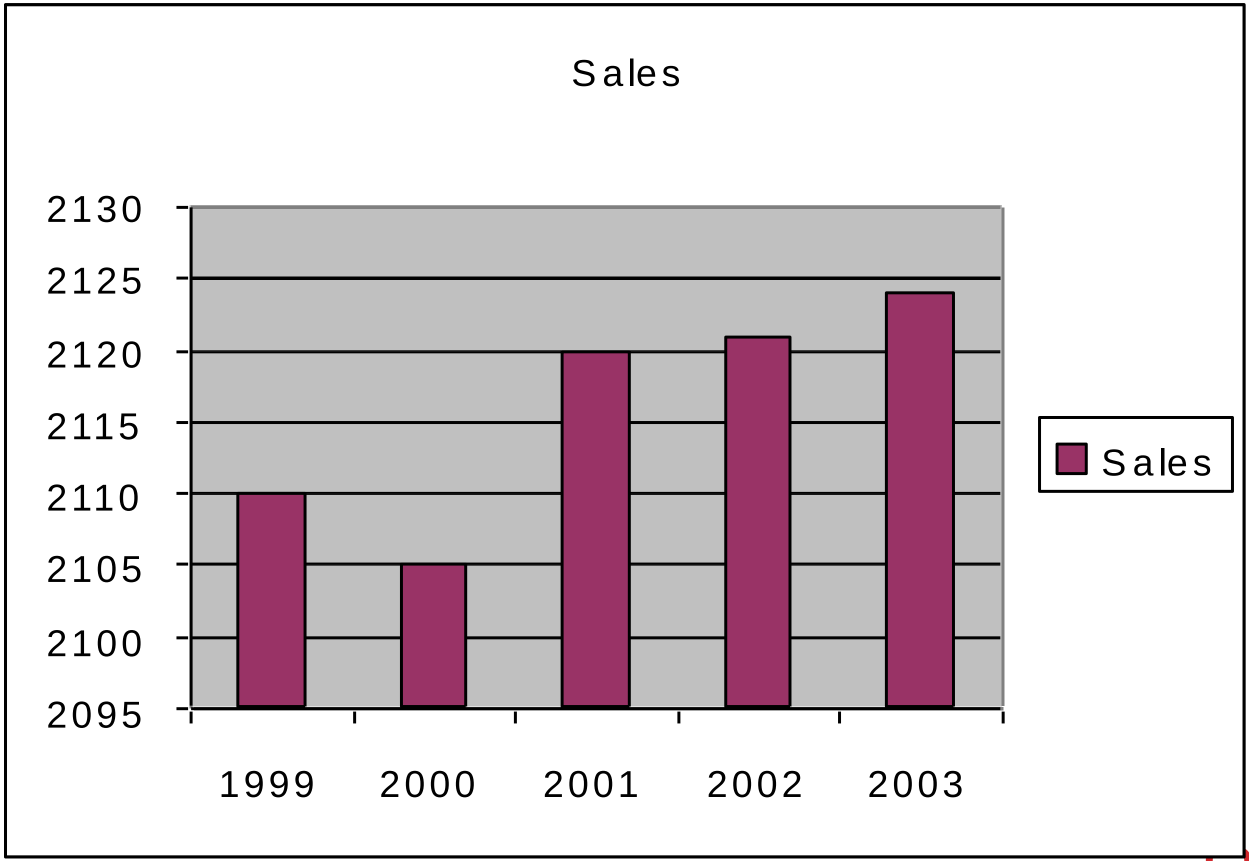
Most graphs become disorderly when data set is very large since it requires human eyes

Be careful regarding false conclusions, can be misleading

BAD VISUALIZATION

Y-Axis scale gives wrong impression of big change.

Year	Sales
1999	2110
2000	2105
2001	2120
2002	2121
2003	2124



BAD VISUALIZATION

***Better Visualization* can be done by scaling the axis from 0 to 2000,**

- It gives correct impression of small change

Year	Sales
1999	2110
2000	2105
2001	2120
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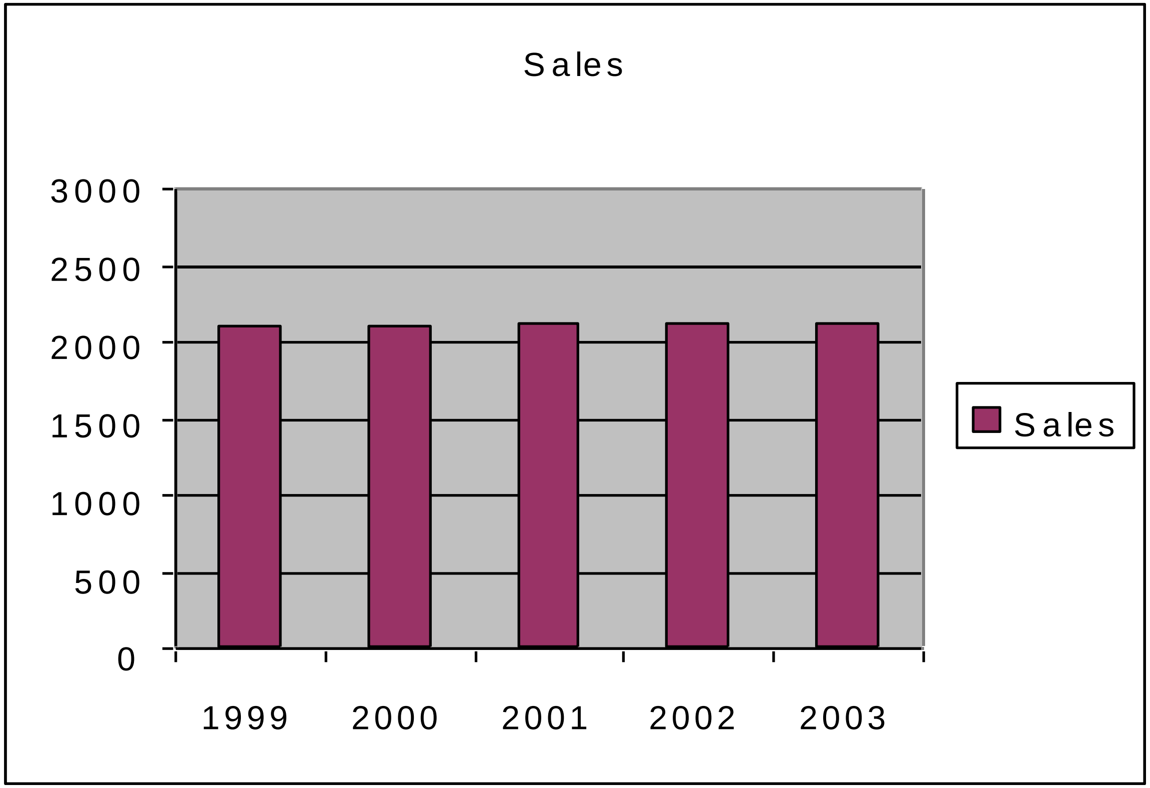
2003	2124



DATA VISUALIZATION - HISTOGRAM

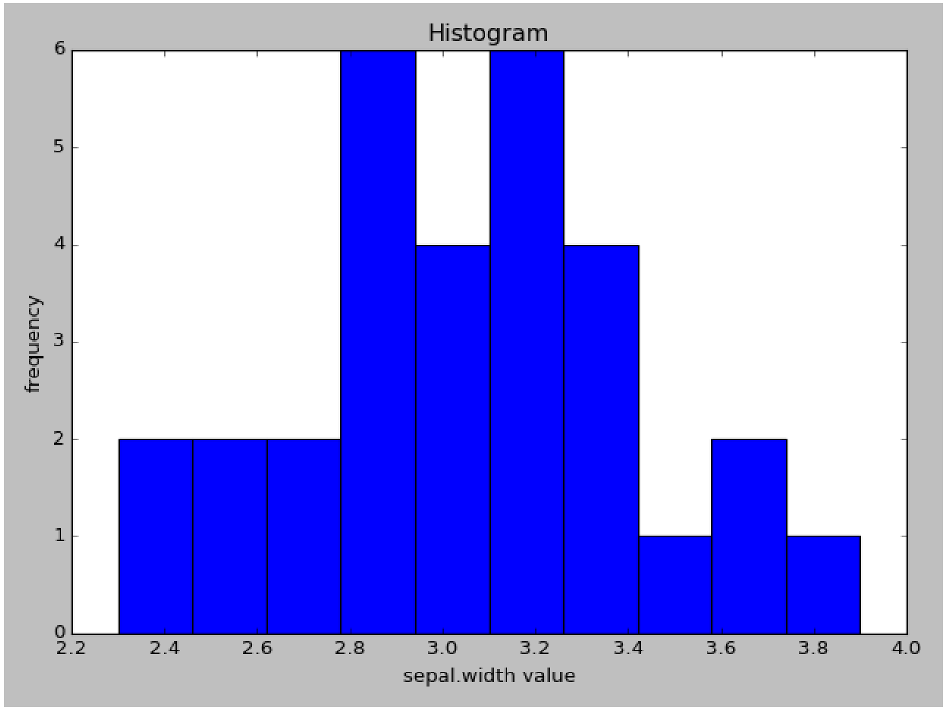
```python
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv("iris.csv")
att = data['sepal.width']

plt.style.use('classic')
plt.hist(att)

plt.title("Histogram")
plt.xlabel("sepal.width value")
plt.ylabel("frequency")

plt.show()
```



DATA VISUALIZATION – PIE CHART

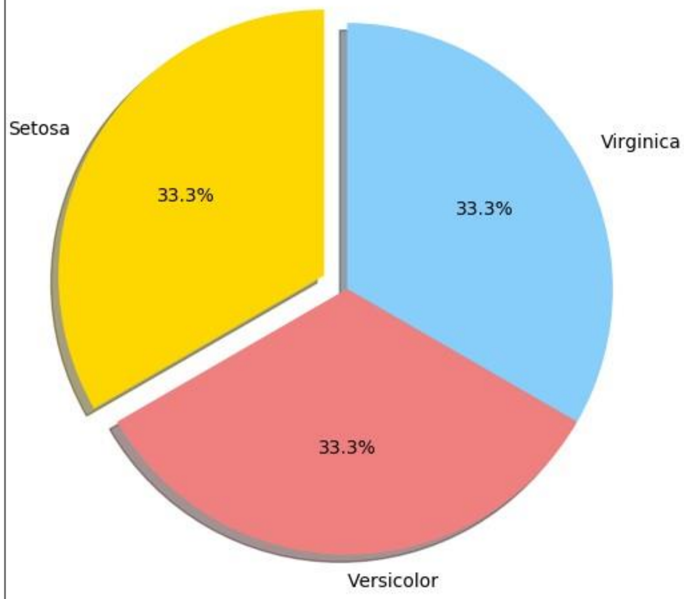
```
import pandas as pd
import matplotlib.pyplot as plt
from collections import Counter

data = pd.read_csv("iris.csv")
att = data['variety']

labels = Counter(att).keys()
counts = Counter(att).values()
colors = ['gold', 'lightcoral', 'lightskyblue']
explode = (0.1, 0, 0)   # explode 1st slice

# Plot
plt.pie(counts, labels=labels,
        explode=explode, colors=colors,
        autopct='%1.1f%%', shadow=True,
        startangle=90)

plt.axis('equal')
plt.show()
```



DATA VISUALIZATION – SCATTER PLOT

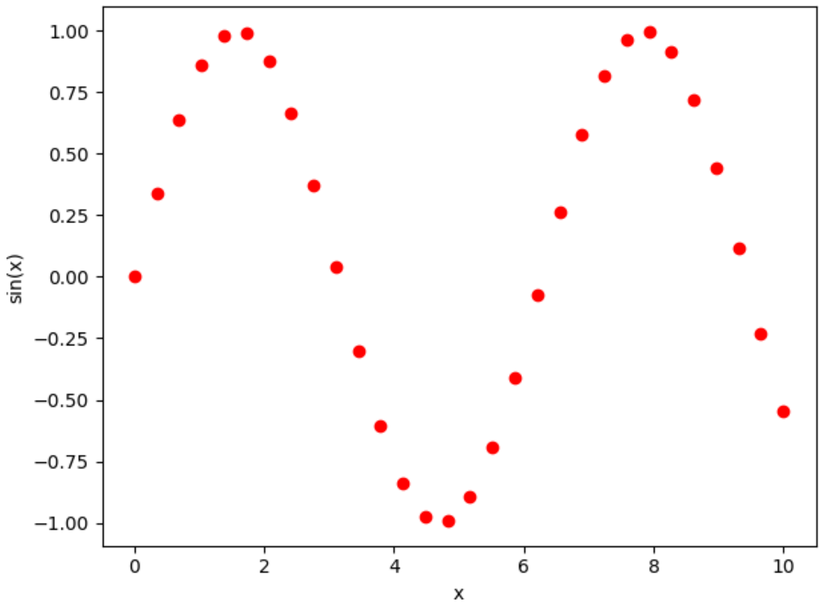
```
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(0, 10, 30)
y = np.sin(x)

plt.xlabel("x")
plt.ylabel("sin(x)")

plt.plot(x, y, 'o', color='red’)

plt.show()
```



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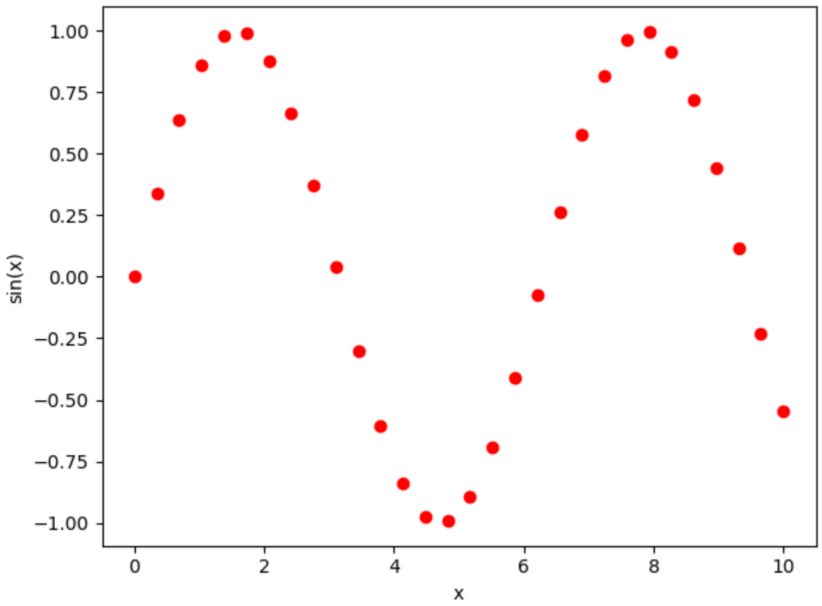


DATA VISUALIZATION – PIE CHART

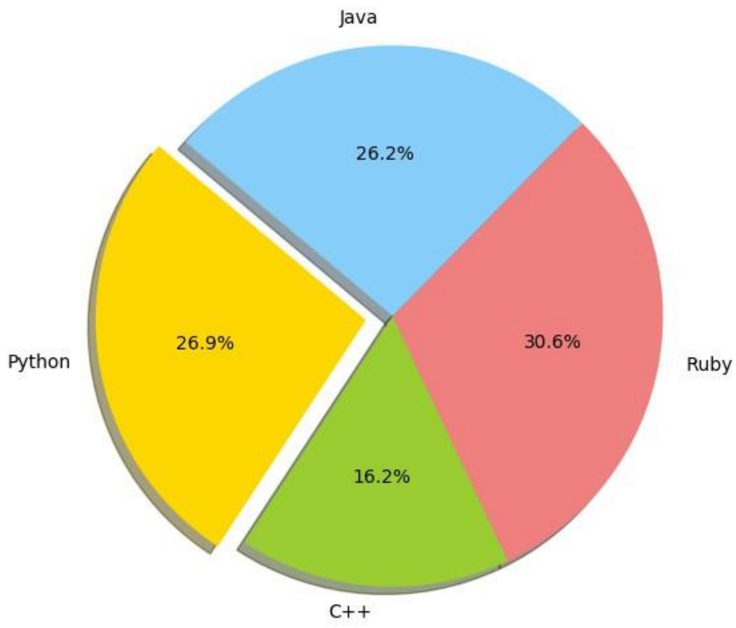
```
import matplotlib.pyplot as plt

# Data to plot consists of labels, sizes and colors for
each # label
labels = 'Python', 'C++', 'Ruby', 'Java'
sizes = [215, 130, 245, 210]
colors = ['gold', 'yellowgreen', 'lightcoral',
'lightskyblue']
explode = (0.1, 0, 0, 0) # explode 1st slice

# Plot
plt.pie(sizes, explode=explode, labels=labels,
        colors=colors, autopct='%1.1f%%',
        shadow=True, startangle=140)

plt.axis('equal')

plt.show()
```



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each # label
labels = 'Python', 'C++', 'Ruby', 'Java'
sizes = [215, 130, 245, 210]
colors = ['gold', 'yellowgreen', 'lightcoral',
'lightskyblue']

# Plot
patches, texts = plt.pie(sizes, colors=colors,
shadow=True,
                    startangle=90)

plt.legend(patches, labels, loc="best")
plt.axis('equal')
```

```
plt.show()
```